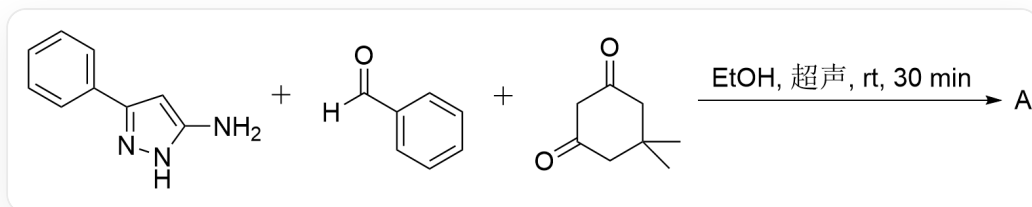


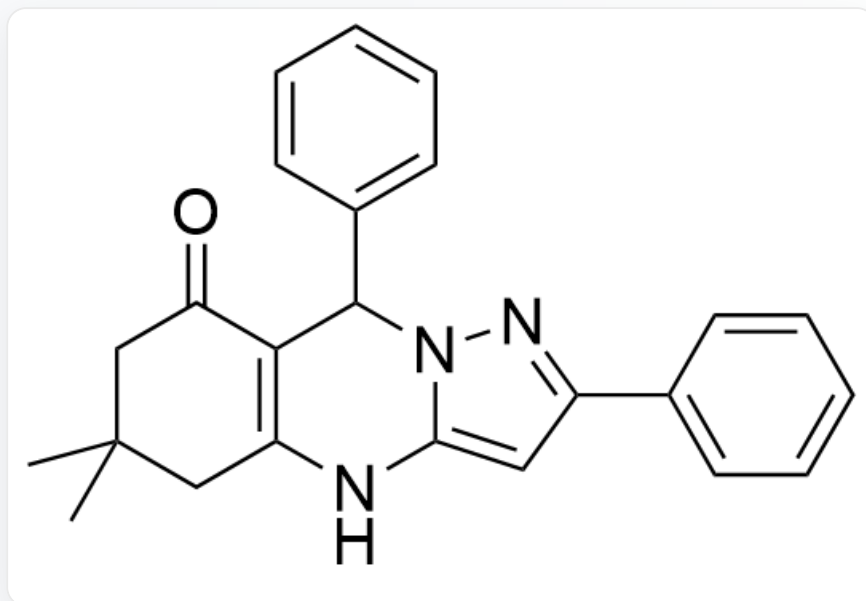
Question



NC1=CC(C2=CC=CC=C2)=NN1.O=C(C1=CC=CC=C1)[H].O=C1CC(C)(C)CC(C1)=O>>[A], reaction conditions: solvent CCO (*EtOH*), ultrasonication, rt, 30 min

Please select the structural formula of product **A** and determine whether it has enantiomers.

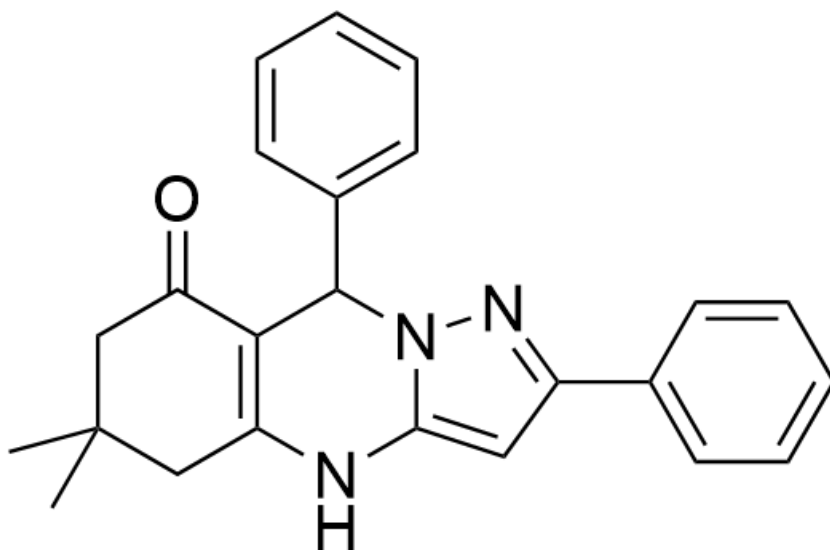
A.



O=C1C2=C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)CC(C)(C)C1

Has enantiomers

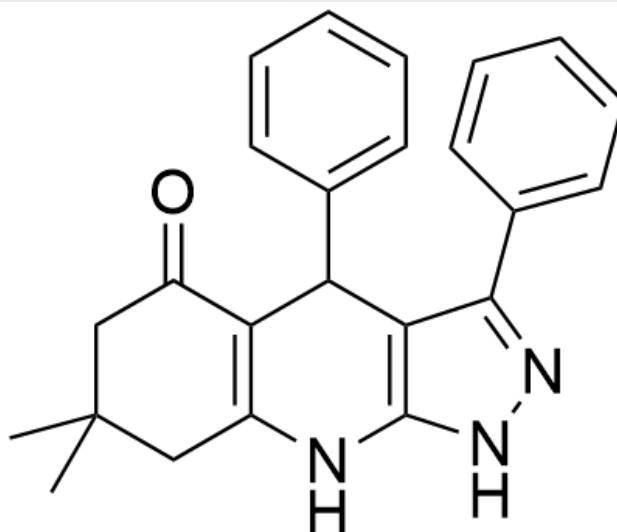
B.



O=C1C2=C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)CC(C)(C)C1

No enantiomer

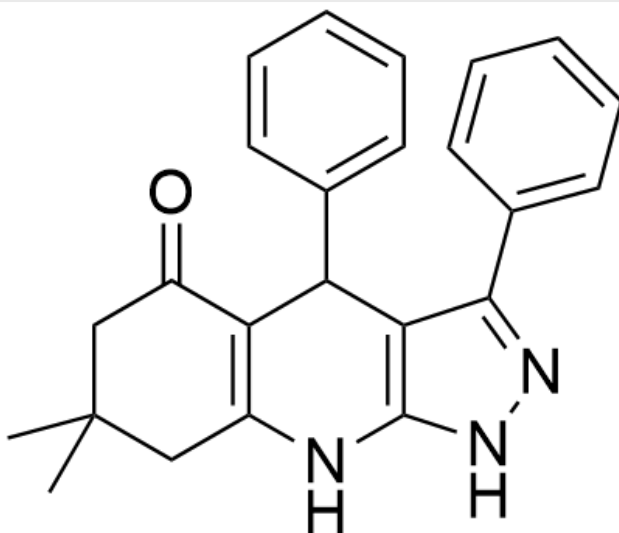
C.



O=C1C2=C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)CC(C)(C)C1

Has enantiomers

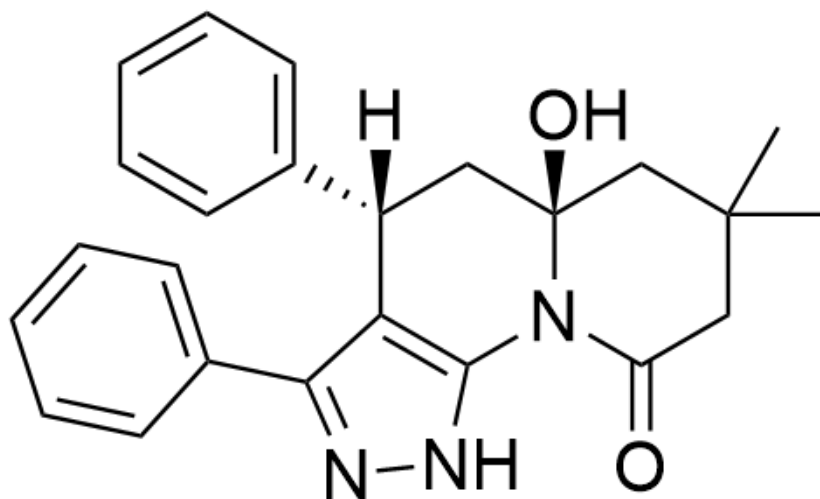
D.



O=C1C2=C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)CC(C)(C)C1

No enantiomer

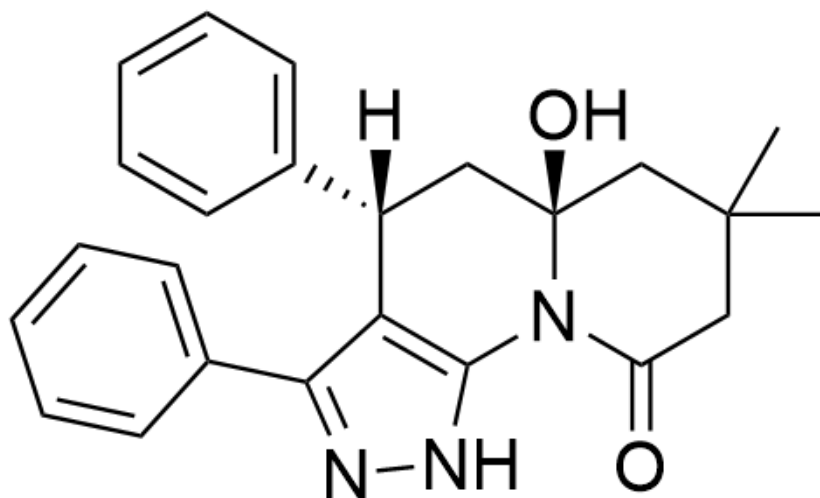
E.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

Has enantiomers

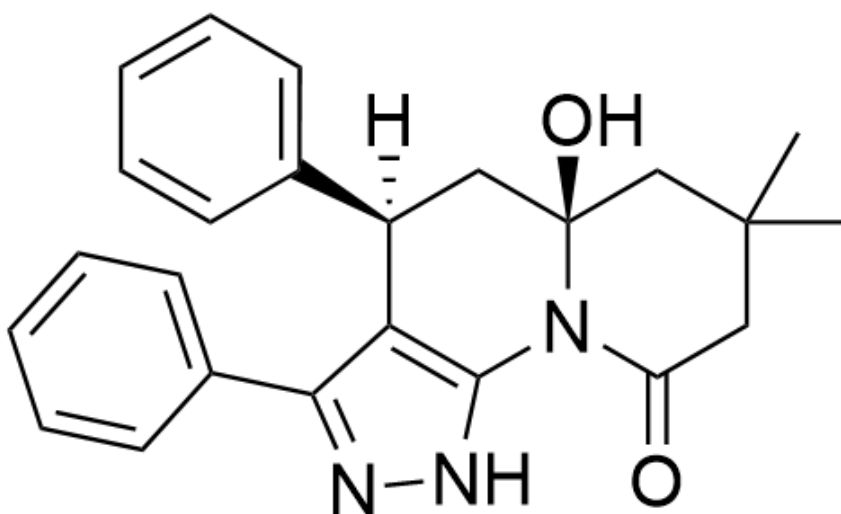
F.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

No enantiomer

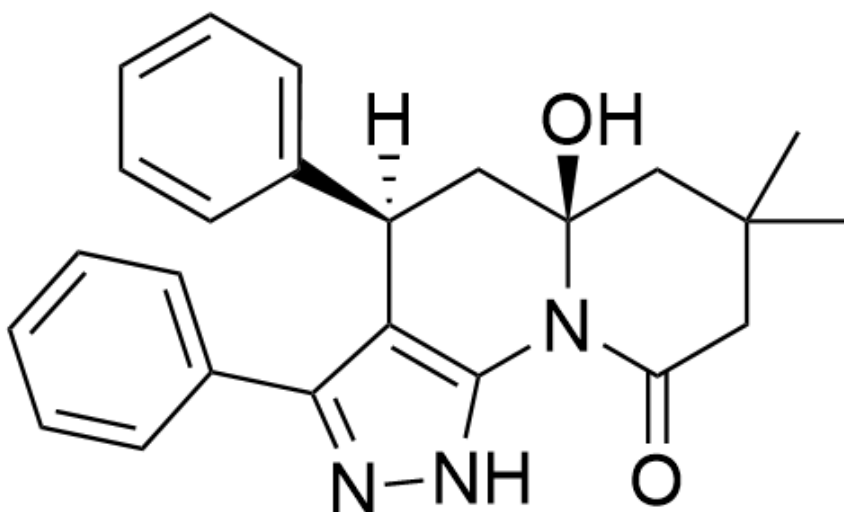
G.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

Has enantiomers

H.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

No enantiomer

Answer

Correct Answer: **A**

Detailed Explanation

Under ambient and neutral conditions, the reaction is kinetically controlled

CHECKPOINT

1 PTS

Under ambient and neutral conditions, the reaction is kinetically controlled

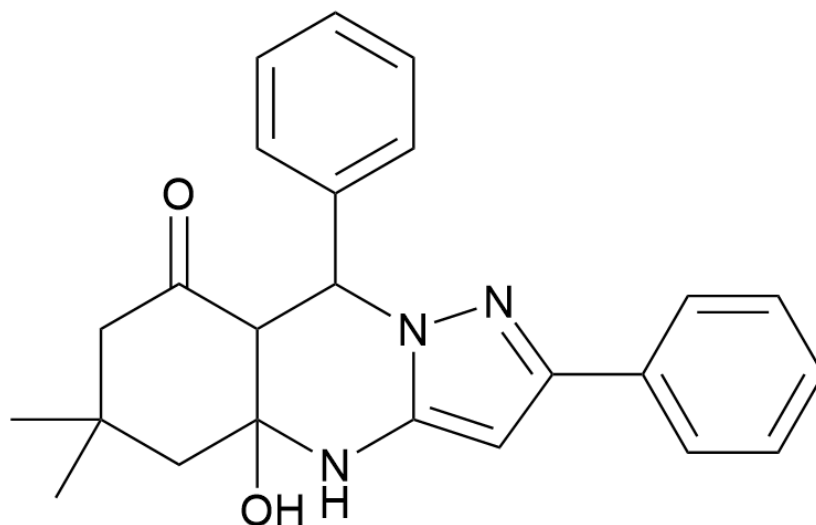
The nucleophilic ability of the nitrogen atom's n-electrons is stronger than that of the π -electrons in a carbon-carbon double bond

CHECKPOINT

1 PTS

The nucleophilic ability of the nitrogen atom's n-electrons is stronger than that of the π -electrons in a carbon-carbon double bond

First, a condensation reaction occurs via a mechanism similar to the Biginelli reaction to yield the intermediate

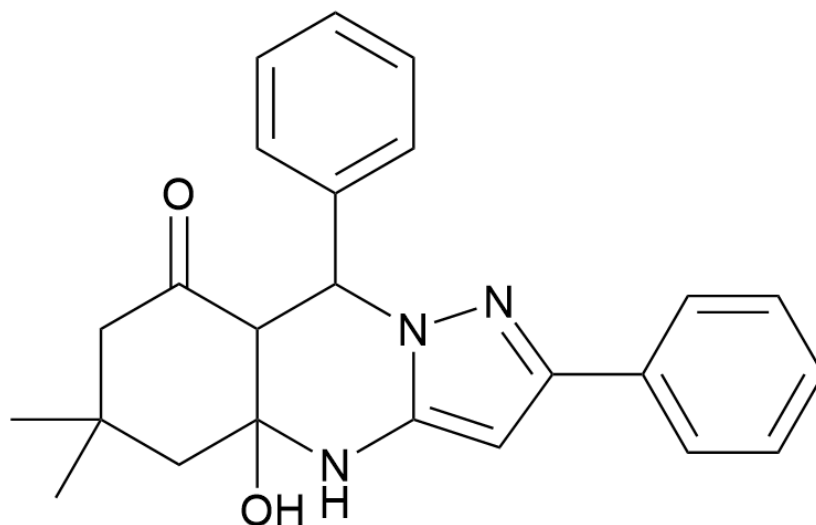


O=C1C2C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)(O)CC(C)(C)C1

CHECKPOINT

1 PTS

First, a condensation reaction occurs via a mechanism similar to the Biginelli reaction to yield the intermediate



O=C1C2C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)(O)CC(C)(C)C1

Subsequently, an elimination reaction occurs to remove a molecule of H_2O , yielding product **A**